STATISTICAL ANALYSIS OF JAZZ IN THE COMPUTER AGE: AN INVESTIGATION OF GRANT GREEN'S IMPROVISATIONAL STYLE

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ABSTRACT

This study investigates the improvisations of jazz guitarist Grant Green using a computer-aided statistical approach. The analysis of improvised jazz has been employed in the transmission of jazz language since its early beginnings, and is still an integral part of the pedagogical process at music institutions. However, the methods used to undertake analysis have not changed substantially in the last 50 years. Brought on by increasing computational power, the computer-aided statistical analysis of jazz is a field of study experiencing considerable growth. This study uses software tools and methods developed within the Jazzomat Research Project to aid in the analysis of the improvisational style of Grant Green, focusing on the years 1960-1965. Grant Green was a prolific but under-appreciated musician during the 1960s and 1970s, and was the unofficial house guitarist for Blue Note records from 1960 to 1965. The data used for analysis will be drawn from 40 detailed transcriptions of Green's improvisations from this period of time.

This research aims to create a statistically rigorous and detailed model of Green's improvisational style with respect to a wide range of independent and dependent variables. The main dependent variables will be features of melodic and rhythmic construction. Melodic construction includes: distribution of pitch classes compared to the key of the song or chord of the moment; phrase shape; and interval size; while rhythmic construction includes: note duration; beat placement; note density; and micro-timing features such as swing ratio. The independent variables include: year of recording; time signature; key signature; the chord of the moment; tempo; section of the form; and location in the solo. Some features, such as solo length and phrase length, will be investigated as both independent and dependent variables. The analysis will also investigate whether there are distinctive patterns that Green uses, and how these may be influenced by the independent variables.

The analysis will be undertaken in multiple steps, including: descriptive statistics and univariate analysis to investigate the features of the dataset, and limit the inclusion of variables in further analysis; bivariate analysis to examine the relationships between pairs of variables, informed by prior studies; and multivariate analysis to study more complex relationships to investigate how Green's improvisational style changes under differing conditions. Due to the variety of data types present in the dataset a range of analytical tests will be employed, including analysis of variance, regression, and compositional analysis.

It is intended that the results of this research will be able to be applied in future research, including machine learning classification and generation tasks, while also extending our understanding of music and approaches to analysing improvised jazz.